Bar Plots

College of Charleston

September 1, 2026

Motivation: Tips Data

First 20 observations of the tips given to a waiter over the course of several months in a restaurant. Do more customers come to the restaurant on certain days?

Total Bill	Tip	Sex	Smoker	Day	Time	Size
13.42	1.58	Male	Yes	Fri	Lunch	2
16.27	2.50	Female	Yes	Fri	Lunch	2
10.09	2.00	Female	Yes	Fri	Lunch	2
20.45	3.00	Male	No	Sat	Dinner	4
13.28	2.72	Male	No	Sat	Dinner	2
22.12	2.88	Female	Yes	Sat	Dinner	2
24.01	2.00	Male	Yes	Sat	Dinner	4
15.69	3.00	Male	Yes	Sat	Dinner	3
11.61	3.39	Male	No	Sat	Dinner	2
10.77	1.47	Male	No	Sat	Dinner	2
15.53	3.00	Male	Yes	Sat	Dinner	2
10.07	1.25	Male	No	Sat	Dinner	2
12.60	1.00	Male	Yes	Sat	Dinner	2
32.83	1.17	Male	Yes	Sat	Dinner	2
35.83	4.67	Female	No	Sat	Dinner	3
29.03	5.92	Male	No	Sat	Dinner	3
27.18	2.00	Female	Yes	Sat	Dinner	2
22.67	2.00	Male	Yes	Sat	Dinner	2
17.82	1.75	Male	No	Sat	Dinner	2
18.78	3.00	Female	No	Thur	Dinner	2

We need something more than a data file to answer that

Claim

Most statistical analysis can be done graphically; the math is high-end filler

- If there is a relationship, it's usually visual
- If there isn't a relationship, it's usually visual
- Trick is finding the right graphics
- If I don't see the data somewhere in the paper I'm suspicious
  - And yes, it's easy to lie with graphics
- I'm basing this off of my time doing consulting work during grad school

Data Visualization

Why do we graph data?

- It (hopefully) allows us to interpret data...
  - quickly and
  - easily

Which graph is chosen is driven by

- the type of data
- the number of variables and
- what we are trying to convey (context)

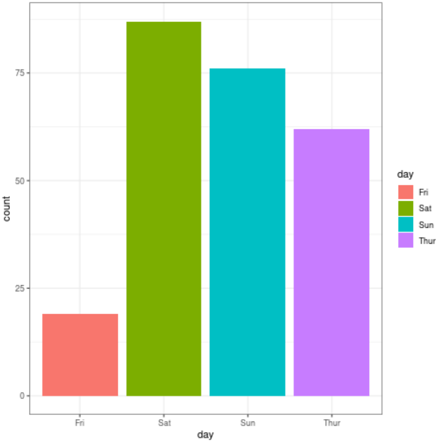
Context

Context gives us the goal of what we want to convey from our data

- Eg If we are interested in the fastest 100m dashes in the Olympics over time we could plot the shortest time for each year
  - three numeric variables (all run times, shortest run times, and years)
  - The context lets us ignore "all run times"
  - More parsimonious graphs
- Often, for a single variable, we are interested in the **distribution**
  - The distribution of a variable is the frequency certain values occur
  - Eg the distribution of of sizes of penguin colonies
  - Heavily tied to probability

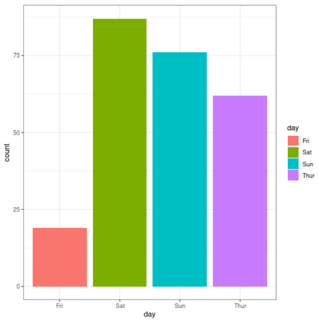
One Categorical Variable: Graph

When we have one categorical variable, a **bar chart** is often used to tally the frequencies (counts) of that categorical variable



One Categorical Variable: Distribution

To describe the distribution of a categorical variable we need to talk about....



- how likely each category is
- the most and least likely category
- any interesting patterns
  - WARNING: The order of a bar chart is sometimes meaningless so it can't be read left to right
- numbers help

When should we use pie charts?

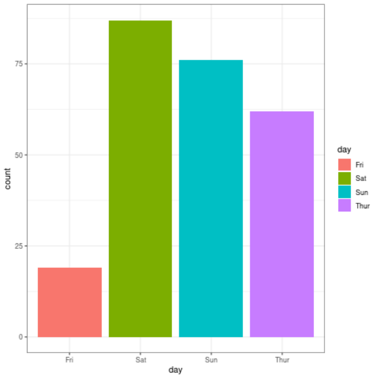
When do we use pie charts?

......never?

- Humans are bad are reading angles/pie slices
- Bar charts can convey the same info, easier
- Dimensions of the graph are weird (bar chart with polar coordinates?)
- No strong advantage to pie charts in my opinion

Counterpoint: Use a table

As a counterpoint to my emphasis on graphics the bar chart (with few categories) has such a poor data-ink ratio that a small table would suffice



- **data-ink ratio**: The relevant amounts of ink/pixels used to convey information vs the amount of ink/pixels used (roughly)
- High data ink ratio is parsimonious and spartan (in a good way)
- Low data ink ratio takes up space and doesn't add much

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Fri	Sat	Sun	Thur
19	87	76	62

Table: Total Days Worked